

Unit 5: Inverse Functions

Notes & Practice

Name: _____

Date: _____

Period: _____

Finding the Inverse of a Square Root Function**What is an Inverse Function?**

An inverse function “undoes” what the original function does. If $f(x)$ takes x to y , then $f^{-1}(y)$ takes y back to x .

Important: The notation f^{-1} does NOT mean $\frac{1}{f}$. It represents the inverse function.

Process for Finding an Inverse:

Step 1: Replace $f(x)$ with y

Step 2: Switch x and y

Step 3: Solve for y (isolate the radical, then square both sides)

Step 4: Replace y with $f^{-1}(x)$

Example 1: Find the inverse function.

A. $f(x) = (x + 3)^{1/2}$

D. $f(x) = \sqrt{x} + 4$

B. $f(x) = \sqrt{x - 5}$

E. $f(x) = 2(x)^{1/2}$

C. $f(x) = (2x + 1)^{1/2}$

Verifying Inverse Functions**How to Verify Two Functions are Inverses:**

Two functions f and g are inverses if and only if:

$$f(g(x)) = x \quad \text{AND} \quad g(f(x)) = x$$

You must check **BOTH** compositions!

Process for Verifying Inverses:

Step 1: Find $f(g(x))$ by substituting $g(x)$ into $f(x)$. Simplify — you should get x .

Step 2: Find $g(f(x))$ by substituting $f(x)$ into $g(x)$. Simplify — you should get x .

Step 3: If **both** equal x , the functions are inverses!

Example 3: Verify that f and g are inverse functions.

A. $f(x) = (x + 5)^{1/2}$ and $g(x) = x^2 - 5$ for $x \geq 0$

B. $f(x) = \sqrt{x + 3}$ and $g(x) = (x - 3)^2$ for $x \geq 3$

Example 4: Verify that f and g are inverse functions.

A. $f(x) = (2x - 4)^{1/2}$ and $g(x) = \frac{x^2 + 4}{2}$ for $x \geq 0$

B. $f(x) = 2\sqrt{x}$ and $g(x) = \frac{x^2}{4}$ for $x \geq 0$

Finding & Verifying Inverses Together**Complete Process:**

1. Find the inverse using the 4-step process
2. Verify by checking that $f(f^{-1}(x)) = x$ AND $f^{-1}(f(x)) = x$

Important Note about Domains:

When finding the inverse of a square root function, the domain of the inverse may be restricted.

- If $f(x) = x^{1/2}$, then $f^{-1}(x) = x^2$ for $x \geq 0$ (domain restriction!)
- The **range** of f becomes the **domain** of f^{-1}
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Example 5: Find the inverse of $f(x)$, then verify your answer.

A. $f(x) = (x + 2)^{1/2}$

B. $f(x) = \sqrt{3x - 6}$

Example 6: Find the inverse and state any domain restrictions.

A. $f(x) = (x - 4)^{1/2}$ for $x \geq 4$

B. $f(x) = \sqrt{x} + 1$ for $x \geq 0$